

Erik Bloomquist

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EDUCATION

Bachelor of Science in Electrical Engineering
University of Florida

Spring 2026
GPA: 3.98/4

WORK EXPERIENCE

Undergraduate Teaching Assistant | *UF ECE Dept.*

August 2024 - Present

- Assists Dr. Catia Silva in operation of Fundamentals of Machine Learning
- Course topics include Bayesian Learning, Generative Models, Clustering, Discriminative Classification, Dimensionality Reduction, Manifold Learning, Neural Networks, and Deep Learning

Undergraduate Teaching Assistant | *UF Engineering Education Dept.*

May 2023 - July 2024

- Assisted Dr. Lilianny Virguez in operation of Elements of Electrical Engineering
- Course topics include AC & DC circuit analysis, Arduino projects, and basic C++ programming

Student Assistant- Database | *UF College of Dentistry*

May 2023 - September 2023

- Worked with the UF College of Dentistry to migrate 5000 medical records into the College's database

RESEARCH EXPERIENCE

Machine Learning for Financial Markets* | *Dr. Catia Silva*

- Designing machine learning frameworks for financial market prediction, focusing on risk modeling and interpretability; serving as project leader and sole student author
- Work supported by grant funding from UF Center for Undergraduate Research's AI Scholars Program

Literature Review: Adversarial Machine Learning* | *Dr. Damon Woodard*

- Surveying adversarial ML attacks and defenses to assess effectiveness and guide future research in robust deep learning
- Work supported by the Florida Institute for National Security (FINS)

From Black-Box to Glass-Box Models: Towards Explainability in Reinforcement Learning | *Dr. Damon Woodard*

- Trained common RL algorithms (DQN, PPO) on benchmark environments (Cart Pole, Mountain Car, Lunar Lander)
- Examined decision boundaries and extracted interpretable policy representations

AWARDS

Dr. Joseph S. Rosko Award | *Scholarship*

Fall 2025

- Awarded \$5000 scholarship for academic achievement and involvement in the ECE department

AI Scholars Program | *Research Funding*

Fall 2025

- Awarded \$1750 grant to lead independent research bridging machine learning, risk modeling, and financial prediction

Florida Institute for National Security (FINS) Talent Pipeline | *Fellowship*

Spring 2024

- Awarded \$5000 fellowship to perform research at the intersection of AI and national security

University Honors Program | *Recognition*

Fall 2022

- Admitted to UF's University Honors Program, providing access to advanced coursework and academic enrichment

PRESENTATIONS

2026 Spring Undergraduate Research Symposium* | *Poster Presentation*

Spring 2026

- Will present my AI Scholars Program work on self-directed research in machine learning for financial market prediction

2025 Spring Undergraduate Research Symposium | *Poster Presentation*

Spring 2025

- Accepted to present poster on interpretable reinforcement learning at the 2025 UF Spring Symposium.

LEADERSHIP & INVOLVEMENT

Undergraduate Curriculum Chair | *Eta Kappa Nu (HKN) Epsilon Sigma*

Spring 2024 - Present

- Represents IEEE's honor society at faculty curriculum committee meetings
- Selected to serve on the ECE student panel, answering curriculum-related questions from underclassmen in electrical and computer engineering

* In progress

RELEVANT COURSEWORK

Machine Learning Mathematics	Applied Machine Learning Systems
Formal Methods for Robotics & AI	Data Science
Stochastic Methods	Communication Systems
Digital Signal Processing	Biomedical Image Analysis
Analog Circuitry	Electrophysics

SKILLS

Programming & Frameworks: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), MATLAB, C++
Data & Tools: Git, Altium Designer, LTspice, LaTeX, Jupyter, SQL
Machine Learning Expertise: Deep Learning, Reinforcement Learning, Adversarial ML, End-to-End ML Systems, Data Cleaning & Preprocessing
Signals & Systems Knowledge: Digital Signal Processing, Biomedical Signal Analysis, Circuit Modeling
Professional Skills: Technical Writing, Effective Communication, Poster Presentations, Teaching & Mentorship